

Chapter-7: Discrete Fourier Transform (DFT)

❑ Discrete-Fourier Transform (DFT) Representation, Properties of DFT:

1. Introduction:

- Frequency analysis of discrete-time signals is usually performed on a ***digital signal processor***, which may be a general-purpose digital computer or specially designed digital hardware.
- To perform frequency analysis on a discrete-time signal $x[n]$, we convert $x[n]$ to an equivalent frequency-domain representation. We know that such a representation is given by the Fourier transform $X(e^{j\omega})$ of the sequence $\{x[n]\}$.
- However, $X(e^{j\omega})$ is continuous function of frequency (ω) and it is not a computationally convenient representation of the sequence $\{x[n]\}$.

- If we represent the sequence $\{x[n]\}$ by samples of its spectrum $X(e^{j\omega})$, it leads to the Discrete Fourier Transform (DFT) which is a powerful computational tool for performing frequency analysis of discrete-time signals.
- The DFT is applicable to finite-length sequence.
- The ***DFT is itself a sequence rather than a function of a continuous variable***, and it corresponds to samples, equally spaced in frequency, of the DTFT of the signal.
- The ***DFT plays a central role in the implementation of a variety of digital signal-processing algorithms***. This is because efficient algorithms exist for the computation of the DFT.

2. Definition:

- **DFT (Discrete Fourier Transform) Definition:**

- The ***DFT is defined as a finite length discrete-frequency sequence*** which is obtained by uniformly sampling of Fourier transform $X(e^{j\omega})$. Sampling is done at ' N ' equispaced points over the period $0 \leq \omega \leq 2\pi$.
- Mathematically, the DFT of discrete sequence $x[n]$ is denoted by $X[k]$ and is expressed as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi k}{N}n}, 0 \leq k \leq N-1 \quad \text{.....1}$$

Since, this summation is taken for N points, it is called ***N – point DFT***.

- Equation (1) is also called the ***analysis equation***.

Thus $X(k)$ being complex and has real & imaginary component, so that the k th harmonic of $X(k)$ is:

$$X(k) = R(k) + jI(k)$$

- Here, , k represents the harmonic number of the transform component.

- The DFT can also be written as:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \text{.....2}$$

- Where, $W_N = e^{-j\frac{2\pi}{N}}$, is the **twiddle factor** which makes the computation of DFT a bit easy and fast.

• **IDFT (Inverse Discrete Fourier Transform) Definition:**

- The IDFT is the process of obtaining discrete-time sequence $x[n]$ from the DFT $X[k]$ and is expressed as:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{\frac{j2\pi kn}{N}}, 0 \leq n \leq N-1 \quad \text{.....3}$$

which is called $N - \text{point IDFT}$. Also called **synthesis equation**.

- The IDFT can also be written as:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \quad \text{.....4}$$

3. Comparison Between DTFT and DFT:

DTFT	DFT
➤ The DTFT(Discrete-Time Fourier Transform) is defined as $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$	➤ The DFT (Discrete Fourier Transform) is defined as $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi k}{N}n}, \quad 0 \leq k \leq N-1$
➤ The IDTFT (Inverse Discrete-Time Fourier Transform) is defined as $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$	➤ The IDFT (Inverse Discrete Fourier Transform) is defined as $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi kn}{N}}$
➤ It is infinite length sequence $x[n]$.	➤ It is applicable to finite length sequence $x[n]$.
➤ It is continuous function of frequency ω .	➤ It is obtained by frequency-domain sampling of DTFT.
➤ it is not a computationally convenient representation of the sequence $\{x[n]\}$.	➤ it is computationally convenient representation of the sequence $\{x[n]\}$.

- **Zero Padding:**

- It is the process of adding extra zeros to make/extend the signal up to N —**points**.

Example-1: Find the 4-point DFT of $x(n) = \{1, -1, 2, -2\}$ directly.

Solution:

- Given sequence is $x(n) = \{1, -1, 2, -2\}$. Here, the DFT $X[k]$ to be found is $N = 4$ — **point** and length of the sequence $L = N = 4$. So, no padding of zeros is required. We know that the *DFT* $\{x[n]\}$ is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk} = \sum_{n=0}^{N-1} x(n) e^{-j(2\pi/N)nk} = \sum_{n=0}^3 x(n) e^{-j(\pi/2)nk}, \quad k = 0, 1, 2, 3$$

$$X(0) = \sum_{n=0}^3 x(n) e^0 = x(0) + x(1) + x(2) + x(3) = 1 - 1 + 2 - 2 = 0$$

$$X(1) = \sum_{n=0}^3 x(n) e^{-j(\pi/2)n} = x(0) + x(1) e^{-j(\pi/2)} + x(2) e^{-j\pi} + x(3) e^{-j(3\pi/2)}$$

$$= 1 + (-1)(0 - j) + 2(-1 - j) - 2(0 + j)$$

$$= -1 - j$$

$$X(2) = \sum_{n=0}^3 x(n) e^{-j\pi n} = x(0) + x(1) e^{-j\pi} + x(2) e^{-j2\pi} + x(3) e^{-j3\pi}$$

$$= 1 - 1(-1 - j) + 2(1 - j) - 2(-1 - j) = 6$$

$$X(3) = \sum_{n=0}^3 x(n) e^{-j(3\pi/2)n} = x(0) + x(1) e^{-j(3\pi/2)} + x(2) e^{-j3\pi} + x(3) e^{-j(9\pi/2)}$$

$$= 1 - 1(0 + j) + 2(-1 - j) - 2(0 - j) = -1 + j$$

$$X(k) = \{0, -1 - j, 6, -1 + j\}$$

Example-2: Find the IDFT of $X[k] = \{1, 0, 1, 0\}$.

Solution: (b) Given $X(k) = \{1, 0, 1, 0\}$

Let the IDFT of $X(k)$ be $x(n)$.

$$\therefore x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-nk} = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j(2\pi/N)nk}$$

$$x(0) = \frac{1}{4} \sum_{k=0}^3 X(k) e^0 = \frac{1}{4} [X(0) + X(1) + X(2) + X(3)] = \frac{1}{4} [1 + 0 + 1 + 0] = 0.5$$

$$\begin{aligned} x(1) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j(\pi/2)k} = \frac{1}{4} [X(0) + X(1)e^{j(\pi/2)} + X(2)e^{j\pi} + X(3)e^{j(3\pi/2)}] \\ &= \frac{1}{4} [1 + 0 + e^{j\pi} + 0] = \frac{1}{4} [1 + 0 - 1 + 0] = 0 \end{aligned}$$

$$\begin{aligned} x(2) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j\pi k} = \frac{1}{4} [X(0) + X(1)e^{j\pi} + X(2)e^{j2\pi} + X(3)e^{j3\pi}] \\ &= \frac{1}{4} [1 + 0 + e^{j2\pi} + 0] = \frac{1}{4} [1 + 0 + 1 + 0] = 0.5 \end{aligned}$$

$$\begin{aligned} x(3) &= \frac{1}{4} \sum_{k=0}^3 X(k) e^{j(3\pi/2)k} = \frac{1}{4} [X(0) + X(1)e^{j(3\pi/2)} + X(2)e^{j3\pi} + X(3)e^{j(9\pi/2)}] \\ &= \frac{1}{4} [1 + 0 + e^{j3\pi} + 0] = \frac{1}{4} [1 + 0 - 1 + 0] = 0 \end{aligned}$$

The IDFT of $X(k) = \{1, 0, 1, 0\}$ is $x(n) = \{0.5, 0, 0.5, 0\}$.

4. DFT as Linear Transformation:

➤ The DFT and IDFT is given as

$$DFT\{x[n]\} = X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1$$

and

$$IDFT\{X[k]\} = x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq n \leq N-1$$

➤ The ***N-point DFT*** is expressed ***in matrix form*** as:

$$X_N = W_N x_N \quad \text{.....1}$$

➤ Then the ***IDFT in matrix form*** is given by:

$$x_N = \frac{1}{N} W_N^* X_N \quad \text{.....2}$$

➤ Where, W_N^* is the complex conjugate of W_N . Also,

$$x_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}, X_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix} \text{ and } W_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{N(N-1)} \end{bmatrix}$$

Example: Compute the 4-point DFT of the sequence $x[n] = \{0,1,2,3\}$.

Solution. The first step is to determine the matrix $\mathbf{W}_4$. By exploiting the periodicity property of $\mathbf{W}_4$ and the symmetry property

$$W_N^{k+N/2} = -W_N^k \quad W_N^{k+N} = W_N^k$$

the matrix $\mathbf{W}_4$ may be expressed as

$$\begin{aligned} \mathbf{W}_4 &= \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^1 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^0 & W_4^2 \\ 1 & W_4^3 & W_4^2 & W_4^1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \end{aligned}$$

Then

$$\mathbf{X}_4 = \mathbf{W}_4 \mathbf{x}_4 = \begin{bmatrix} 6 \\ -2 + 2j \\ -2 \\ -2 - 2j \end{bmatrix}$$

- **Example-2:** Compute the 4-point IDFT of the sequence $X[k] = \{6, -2 + 2j, -2, -2 - 2j\}$.

Solution:

- The IDFT is given as, $x_N = \frac{1}{N} W_N^* X_N$
where, W_N^* is the complex conjugate of W_N . Also, $N = 4$.

$$W_N^* = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & +j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & +j \end{bmatrix}$$

- Then, $x_N = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & +j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & +j \end{bmatrix} \begin{bmatrix} 6 \\ -2 + 2j \\ -2 \\ -2 - 2j \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$

- Therefore, $x_N = x[n] = \{0, 1, 2, 3\}$

- **Examples:**

- a. Determine the 4-point DFT of the sequence

$$x[n] = 2u[n + 1] - u[n] + 3u[n - 1] + 2u[n - 2]$$

$$(\text{Ans. } X[k] = \{17, -5 + 2j, -3, -5 - 2j\})$$

- b. Find the 4-point DFT of the sequence

$$x[n] = \delta[n] + 2\delta[n - 1] + 3\delta[n - 2] + 3\delta[n - 3]$$

- c. Find the 4-point DFT of the sequence

$$x[n] = \cos \frac{n\pi}{4}, \quad 0 \leq n \leq 3$$

- d. Find the 4-point DFT of the sequence

$$x[n] = \{1, 2, 3\}$$

□ Properties of DFT:

- The DFT is important to the analysis and design of digital signal processing systems. The properties are described below:

a. Periodicity:

- If $x[n] \xleftrightarrow{DFT} X[k]$
then $x[n + N] = x[n]$ for all n .
 $X[k + N] = X[k]$ for all k .

Proof:

- We know, $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$

then
$$x[n + N] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-k(n+N)}$$

$$x[n + N] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} W_N^{-kN}$$

➤ Or,
$$x[n + N] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} W_N^{-kN}$$

where, $W_N = e^{-j\frac{2\pi}{N}}$, then $W_N^{-kN} = e^{-j\frac{2\pi}{N}N} = e^{-j2\pi} = 1$

➤ Or,
$$x[n + N] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

➤ $\therefore$
$$\mathbf{x[n + N] = x[n]}$$

➤ In similar way, we can prove, $\mathbf{X[k + N] = X[k]}$.

b. Linearity:

➤ If
$$x_1[n] \xleftrightarrow{DFT} X_1[k]$$

and
$$x_2[n] \xleftrightarrow{DFT} X_2[k]$$

then
$$x[n] = ax_1[n] + bx_2[n] \xleftrightarrow{DFT} X[k] = aX_1[k] + bX_2[k]$$

Proof:

➤ We know the definition is defined as

$$DFT\{x[n]\} = X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1$$

where, $x[n] = ax_1[n] + bx_2[n]$

then

$$\begin{aligned} DFT\{x[n]\} &= X[k] = \sum_{n=0}^{N-1} \{ax_1[n] + bx_2[n]\} W_N^{kn} \\ &= a\{\sum_{n=0}^{N-1} x_1[n] W_N^{kn}\} + b\{\sum_{n=0}^{N-1} x_2[n] W_N^{kn}\} \\ &= aX_1[k] + bX_2[k] = X[k] \end{aligned}$$

➤ Therefore,

$$x[n] = ax_1[n] + bx_2[n] \xrightarrow{DFT} X[k] = aX_1[k] + bX_2[k]$$

c. Circular Shift of a Sequence (Or, Circular Symmetries of a Sequence):

- Let $x_p[n]$ be a sequence of period N , which is obtained by periodically extending $x[n]$. That is,

$$x_p[n] = \sum_{l=-\infty}^{\infty} x[n - lN] \quad \text{.....1}$$

- Shifting the periodic sequence $x_p[n]$ by k units to the right,

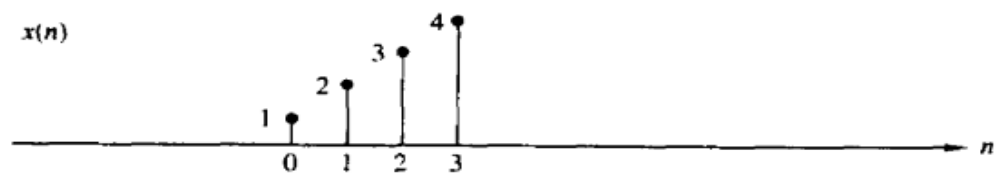
$$x'_p[n] = x_p[n - k] = \sum_{l=-\infty}^{\infty} x[n - k - lN] \quad \text{.....2}$$

- Now, the finite duration sequence

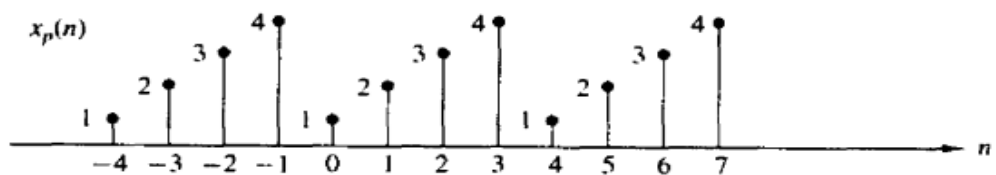
$$x'[n] = \begin{cases} x'_p[n], & 0 \leq n \leq N - 1 \\ 0, & \text{Otherwise} \end{cases} \quad \text{.....3}$$

here, $x'[n]$ is related to the original sequence $x[n]$ by a **circular shift**.

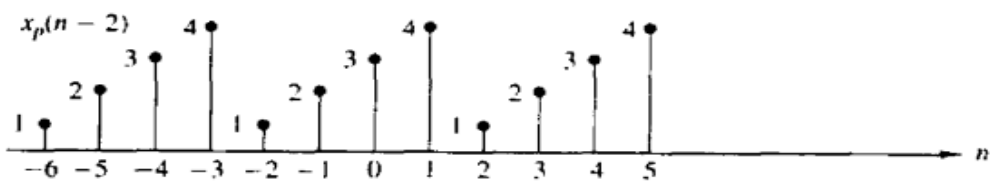
- This relationship is illustrated in figure below for $N = 4$.



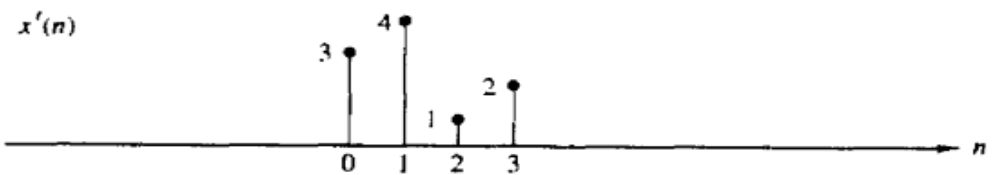
(a)



(b)



(c)



(d)

- **Notation:**

- In general, *the circular shift the sequence can be represented as the index modulo N* . Thus, we can write,

$$x'[n] = x[n - k, \text{modulo } N] = x[(n - k)_N] \quad \dots 4$$

- For example, if $k = 2$ and $N = 4$, we have

$$x'[n] = x[(n - k)_N] = x[(n - 2)_4] \text{ which implies that}$$

$$\begin{aligned} x'[0] &= x[(-2)_4] = x[4 - 2] = x[2] \\ x'[1] &= x[(1 - 2)_4] = x[4 - 1] = x[3] \\ x'[2] &= x[(2 - 2)_4] = x[(0)_4] = x[0] \\ x'[3] &= x[(3 - 2)_4] = x[(1)_4] = x[1] \end{aligned}$$

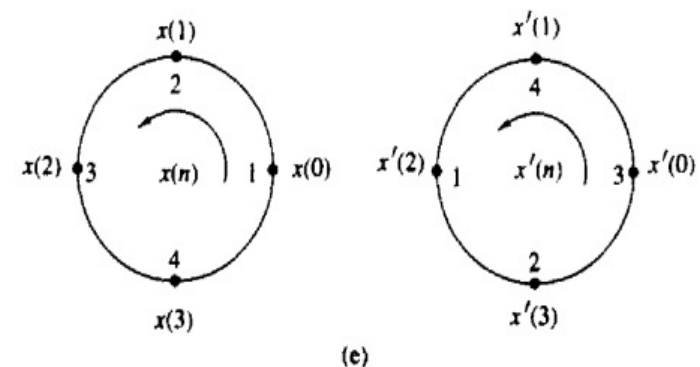


Figure 5.7 Circular shift of a sequence.

- **Cases:**

- i. Circularly Even Sequence:**

- An N –point sequence is called circularly even if it is symmetric about the point zero on the circle. This implies that

$$x[N - n] = x[n], \quad 1 \leq n \leq N - 1 \quad \text{.....5}$$

- ii. Circularly Odd Sequence:**

- An N –point sequence is called circularly odd if it is antisymmetric about the point zero on the circle. This implies that

$$x[N - n] = -x[n], \quad 1 \leq n \leq N - 1 \quad \text{.....6}$$

Examples:

- 1) Check whether the given sequence $x[n] = \{1, 4, 3, 4\}$ is circularly even sequence or not.
- 2) Check whether the given sequence $x[n] = \{2, -3, 0, 3\}$ is circularly odd sequence or not.

Solution:

1. Given, $x[n] = \{1,4,3,4\}$

For circularly even sequence,

$$x[N - n] = x[n], \quad 1 \leq n \leq N - 1, N = 4$$

○ For $n = 1$,

$$x[4 - 1] = x[1] \quad \text{i.e.,} \quad x[3] = x[1] = 4$$

○ For $n = 2$,

$$x[4 - 2] = x[2] \quad \text{i.e.,} \quad x[2] = x[2] = 3$$

○ For $n = 3$,

$$x[4 - 3] = x[3] \quad \text{i.e.,} \quad x[1] = x[3] = 4$$

➤ Since, the given conditions are satisfied all the values of n , the given sequence $x[n] = \{1,4,3,4\}$ is circularly even sequence.

Solution:

2. Given, $x[n] = \{2, -3, 0, 3\}$

For circularly odd sequence,

$$x[N - n] = -x[n], \quad 1 \leq n \leq N - 1, N = 4$$

○ For $n = 1$,

$$x[4 - 1] = -x[1] \quad \text{i.e.,} \quad x[3] = -x[1]$$

○ For $n = 2$,

$$x[4 - 2] = -x[2] \quad \text{i.e.,} \quad x[2] = -x[2]$$

○ For $n = 3$,

$$x[4 - 3] = -x[3] \quad \text{i.e.,} \quad x[1] = -x[3]$$

➤ Since, the given conditions are satisfied all the values of n , the given sequence $x[n] = \{2, -3, 0, 3\}$ is circularly odd sequence.

d. Multiplication of Two DFTs and Circular Convolution:

- We have, the DFT of two N-point sequences $x_1[n]$ and $x_2[n]$ are

$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] e^{-j\frac{2\pi}{N}kn}, \quad 0 \leq k \leq N-1 \quad \text{.....1}$$

and
$$X_2[k] = \sum_{n=0}^{N-1} x_2[n] e^{-j\frac{2\pi}{N}kn}, \quad 0 \leq k \leq N-1 \quad \text{.....2}$$

- Then their product (multiplication of two DFTs)

$$X_3[k] = X_1[k] \times X_2[k], \quad 0 \leq k \leq N-1 \quad \text{.....3}$$

- Now, the IDFT of $X_3[k]$ is

$$x_3[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_3[k] e^{j\frac{2\pi}{N}kn}, \quad 0 \leq n \leq N-1$$

$$x_3[n] = \frac{1}{N} \sum_{k=0}^{N-1} \{X_1[k] \times X_2[k]\} e^{j\frac{2\pi}{N}kn} \quad \text{....4}$$

➤ From equations (1), (2), and (4), we have

$$x_3[n] = \frac{1}{N} \sum_{k=0}^{N-1} \left\{ \sum_{m=0}^{N-1} x_1[m] e^{-j\frac{2\pi}{N}km} \times \sum_{l=0}^{N-1} x_2[l] e^{-j\frac{2\pi}{N}kl} \right\} e^{j\frac{2\pi}{N}kn}$$

$$x_3[n] = \frac{1}{N} \sum_{m=0}^{N-1} x_1[m] \sum_{l=0}^{N-1} x_2[l] \left\{ \sum_{k=0}^{N-1} (e^{j2\pi(n-m-l)/N})^k \right\} \quad \text{.....5}$$

➤ We know,

$$\sum_{k=0}^{N-1} (e^{j2\pi(n-m-l)/N})^k = \begin{cases} N, & n - m - l = pN; l = n - m - pN; l = ((n - m))_N \\ 0, & \text{Otherwise} \end{cases}$$

➤ Then equation (5) becomes

$$x_3[n] = \frac{1}{N} \sum_{m=0}^{N-1} x_1[m] \sum_{l=0}^{N-1} x_2[l] N$$

➤ Therefore, $x_3[n] = \sum_{m=0}^{N-1} x_1[m] x_2[((n - m))_N]$ 6

Equation (6) is called *circular convolution*.

- Circular convolution can also be written as

$$x_3[n] = x_2[n] \circledast x_1[n]$$

- Thus, *the multiplication of two DFTs of two sequences is equivalent to the circular convolution of the two sequences in the time domain.*

- **Example:** Perform the circular convolution of the following two sequences

$$x_1[n] = \{\bar{2}, 1, 2, 1\} \quad \text{and} \quad x_2[n] = \{\bar{1}, 2, 3, 4\}$$

Solution:

1) Mathematical Method:

- The circular convolution is given by the equation

$$x_3[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N], \text{ given } N = 4$$

then,
$$x_3[n] = \sum_{m=0}^3 x_1[m] x_2[(n-m)_4]$$

i. For $n = 0$:

$$x_3[0] = \sum_{m=0}^3 x_1[m] x_2[((-m))_4]$$

$$x_3[0] = x_1[0] x_2[((0))_4] + x_1[1] x_2[((-1))_4] + x_1[2] x_2[((-2))_4] \\ + x_1[3] x_2[((-3))_4]$$

$$x_3[0] = x_1[0] x_2[0] + x_1[1] x_2[4-1] + x_1[2] x_2[4-2] \\ + x_1[3] x_2[4-3]$$

$$x_3[0] = x_1[0] x_2[0] + x_1[1] x_2[3] + x_1[2] x_2[2] \\ + x_1[3] x_2[1] = 2 \times 1 + 1 \times 4 + 2 \times 3 + 1 \times 2 = \mathbf{14}$$

$$x_1[n] = \{\bar{2}, 1, 2, 1\} \text{ and} \\ x_2[n] = \{\bar{1}, 2, 3, 4\}$$

ii. For $n = 1$:

$$x_3[1] = \sum_{m=0}^3 x_1[m] x_2[((1-m))_4]$$

$$x_3[1] = x_1[0] x_2[((1))_4] + x_1[1] x_2[((0))_4] + x_1[2] x_2[((1-2))_4] \\ + x_1[3] x_2[((1-3))_4]$$

$$x_3[1] = x_1[0] x_2[((1))_4] + x_1[1] x_2[((0))_4] + x_1[2] x_2[((-1))_4] \\ + x_1[3] x_2[((-2))_4]$$

$$x_3[1] = x_1[0] x_2\left[\left((1)\right)_4\right] + x_1[1] x_2\left[\left((0)\right)_4\right] + x_1[2] x_2[4-1] \\ + x_1[3] x_2[4-2]$$

$$x_3[1] = x_1[0] x_2[1] + x_1[1] x_2[0] + x_1[2] x_2[3] \\ + x_1[3] x_2[2] = 2 \times 2 + 1 \times 1 + 2 \times 4 + 1 \times 3 = \mathbf{16}$$

iii. For $n = 2$: $x_3[2] = \sum_{m=0}^3 x_1[m] x_2\left[\left((2-m)\right)_4\right]$

$$x_3[2] = x_1[0] x_2\left[\left((2)\right)_4\right] + x_1[1] x_2\left[\left((1)\right)_4\right] + x_1[2] x_2\left[\left((0)\right)_4\right] \\ + x_1[3] x_2\left[\left((2-3)\right)_4\right]$$

$$x_3[2] = x_1[0] x_2[2] + x_1[1] x_2[1] + x_1[2] x_2[0] \\ + x_1[3] x_2\left[\left((-1)\right)_4\right]$$

$$x_3[2] = x_1[0] x_2[2] + x_1[1] x_2[1] + x_1[2] x_2[0] \\ + x_1[3] x_2[3] = 2 \times 3 + 1 \times 2 + 2 \times 1 + 1 \times 4 = \mathbf{14}$$

$$x_1[n] = \{\bar{2}, 1, 2, 1\} \text{ and} \\ x_2[n] = \{\bar{1}, 2, 3, 4\}$$

iv. For $n = 3$:
$$x_3[3] = \sum_{m=0}^3 x_1[m] x_2[(3-m)_4]$$

$$x_3[3] = x_1[0] x_2[(3)_4] + x_1[1] x_2[(2)_4] + x_1[2] x_2[(1)_4] + x_1[3] x_2[(0)_4]$$

$$x_3[3] = x_1[0] x_2[3] + x_1[1] x_2[2] + x_1[2] x_2[1] + x_1[3] x_2[0] = 2 \times 4 + 1 \times 3 + 2 \times 2 + 1 \times 1 = 16$$

➤ Therefore, $x_3(n) = \{14, 16, 14, 16\}$

$$x_1[n] = \{\bar{2}, 1, 2, 1\} \text{ and } x_2[n] = \{\bar{1}, 2, 3, 4\}$$

2) Circle (or Graphical) Method:

- In this method, given sequences are displayed on two circles with the same diameter. The sequences are displayed in a clockwise (or anticlockwise) direction.
- The circles having the values $x_1[m]$ displayed on its circumference is held fixed and the circle with the value $x_2[(n-m)_N]$ displayed on its circumference is rotated so that $x_1[m]$ and $x_2[(n-m)_N]$ are aligned.

➤ As in the linear convolution, ***circular convolution involves four steps:***

- a. Folding one sequence
- b. Circular shifted the folded sequence
- c. Multiplying the two sequences to obtain the two sequences, and
- d. Summing the values of product sequence.

- **Example:** Perform the circular convolution of the following two sequences using ***circle method***.

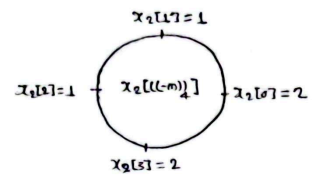
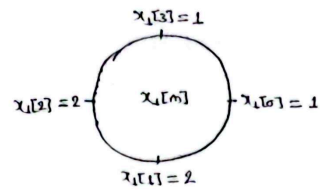
$$x_1[n] = \{1, 2, 2, 1\} \quad \text{and} \quad x_2[n] = \{\bar{2}, 1, 1, 2\}$$

Solution:

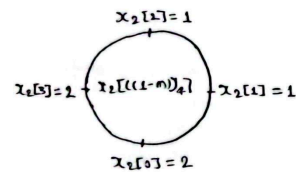
➤ The circular convolution is given as

$$x_3[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N] , \text{ given } N = 4$$

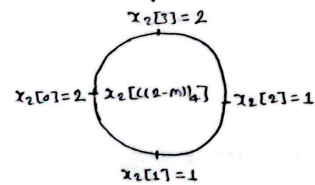
$$x_3[n] = \sum_{m=0}^3 x_1[m] x_2[(n-m)_4]$$



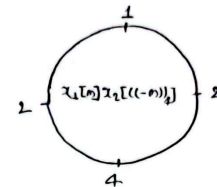
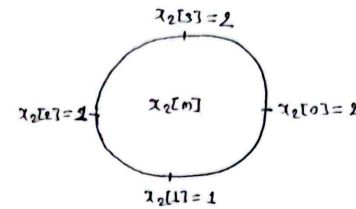
Folded sequence
 $\Downarrow$ rotated by one unit
 in clockwise direction



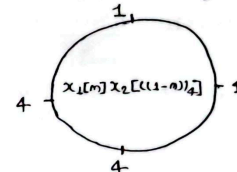
$\Downarrow$ rotated by one unit
 in clockwise direction



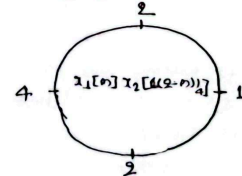
$\Downarrow$ rotated by one unit
 in clockwise direction



Product sequence
 $\therefore x_3[0] = \sum_{m=0}^3 x_1[m] x_2[(1-m)_4]$
 $= 2 + 4 + 2 + 1 = 9$



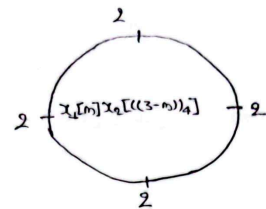
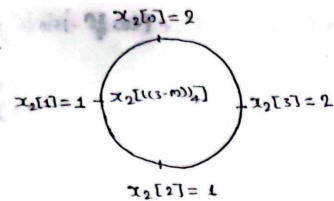
Product sequence
 $\therefore x_3[1] = \sum_{m=0}^3 x_1[m] x_2[(1-m)_4]$
 $= 1 + 4 + 4 + 1 = 10$



Product sequence
 $\therefore x_3[2] = \sum_{m=0}^3 x_1[m] x_2[(2-m)_4]$
 $= 1 + 2 + 4 + 2 = 9$

$$x_3[n] = \sum_{m=0}^3 x_1[m] x_2[(n-m)_4]$$

where, $x_1[n] = \{1, 2, 2, 1\}$ and
 $x_2[n] = \{2, 1, 1, 2\}$



Product sequence

$$\begin{aligned} x_3[3] &= \sum_{m=0}^3 x_1[m] x_2[(3-m)]_4 \\ &= 2 + 2 + 2 + 2 = 8 \end{aligned}$$

$$x_3[n] = \{9, 10, 9, 8\}$$

Method 3: Stockham's Method:

→ Also known as DFT and IDFT method.

$$\begin{aligned} x_3[n] &= \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)]_N \\ &= \text{IDFT} \{ x_1[k] x_2[k] \} \end{aligned}$$

Example: Compute the circular convolution of two sequences given by

$$x_1[n] = \{1, 2, 2, 1\} \quad ; \quad x_2[n] = \{2, 1, 1, 2\}$$

DFT and using IDFT approach.

→ Solution:

→ First we compute $x_1[k]$ and $x_2[k]$. The 4-point DFT

$$x_1[k] = \sum_{n=0}^3 x_1[n] e^{-j\frac{2\pi}{4}kn}, \quad k=0,1,2,3$$

$$x_1[k] = x_1[0] + x_1[1] e^{-j\frac{\pi}{2}k} + x_1[2] e^{-j\pi k} + x_1[3] e^{-j\frac{3\pi}{2}k}$$

$$x_1[k] = 1 + 2 e^{-j\frac{\pi}{2}k} + 2 e^{-j\pi k} + e^{-j\frac{3\pi}{2}k}$$

$$\text{Now, } x_1[0] = 1 + 2 + 2 + 1 = 6$$

$$x_1[1] = 1 + 2(\cos \pi - j \sin \pi) + 2(\cos \pi - j \sin \pi) + (\cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2})$$

$$= 1 - 2j - 2 + j = -1 - j$$

$$x_1[2] = 1 + 2(\cos \pi - j \sin \pi) + 2(\cos 2\pi - j \sin 2\pi) + (\cos 3\pi - j \sin 3\pi)$$

$$= 1 - 2 + 2 - 1 = 0$$

$$x_1[3] = 1 + 2e^{-j\frac{3\pi}{4}} + 2e^{-j\pi} + e^{-j\frac{9\pi}{4}} = 1 + 2j - 2 - j = -1 + j$$

$$\therefore x_1[k] = \{6, -1-j, 0, -1+j\}$$

→ Similarly,

$$x_2[k] = \sum_{n=0}^3 x_2[n] e^{-j\frac{\pi}{2}kn}, \quad k=0,1,2,3$$

$$x_2[k] = x_2[0] + x_2[1] e^{-j\frac{\pi}{2}k} + x_2[2] e^{-j\pi k} + x_2[3] e^{-j\frac{3\pi}{2}k}$$

$$x_2[0] = 2 + e^{-j\frac{\pi}{2}k} + e^{-j\pi k} + 2e^{-j\frac{3\pi}{2}k}$$

Now,

$$x_2[0] = 2 + 1 + 1 + 2 = 6$$

$$x_2[1] = 2 - j - 1 + 2j = 1 + j$$

$$x_2[2] = 2 + e^{-j\pi} + e^{-j2\pi} + 2e^{-j3\pi} = 2 - 1 + 1 - 2 = 0$$

$$x_2[3] = 2 + j - 1 - 2j = 1 - j$$

$$\therefore x_2[k] = \{6, 1+j, 0, 1-j\}$$

$$\rightarrow \text{Now, } x_3[k] = x_1[k] x_2[k]$$

$$\text{then } x_3[0] = x_1[0] x_2[0] = 6 \times 6 = 36$$

$$x_3[1] = x_1[1] x_2[1] = (-1-j)(1+j) = -2j$$

$$x_3[2] = x_1[2] x_2[2] = 0 \times 0 = 0$$

$$x_3[3] = x_1[3] x_2[3] = (-1+j)(1-j) = 2j$$

→ We know that IDFT of $x_3[k]$ is

$$x_3[n] = \frac{1}{4} \sum_{k=0}^3 x_3[k] e^{j\frac{2\pi}{4}kn}$$

$$x_3[n] = \frac{1}{4} [36 + (-2j) e^{j\frac{\pi}{2}n} + 2j e^{j\frac{3\pi}{2}n}]$$

$$x_3[n] = \frac{1}{4} [36 - 2j e^{j\frac{\pi}{2}n} + 2j e^{j\frac{3\pi}{2}n}]$$

$$x_3[0] = \frac{1}{4} [36 - 2j + 2j] = \frac{36}{4} = 9$$

$$x_3[1] = \frac{1}{4} [36 - (2j)j + 2j(-j)] = \frac{1}{4} [36 + 2 + 2] = 10$$

$$x_3[2] = \frac{1}{4} [36 - 2j(-1) + 2j(-1)] = \frac{1}{4} [36 + 2j - 2j] = 9$$

$$x_3[3] = \frac{1}{4} [36 - 2j(-j) + 2j(j)] = \frac{1}{4} [36 - 2 - 2] = 8$$

$$\therefore x_3[n] = \{9, 10, 9, 8\}$$

Note:

* In this method, also we can use the relationships

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \text{and} \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

to calculate DFTs and IDFTs.

Then we can write, for $N=4$

$$X_1[k] = \sum_{n=0}^3 x_1[n] W_4^{kn} = x_1[0] + x_1[1] W_4^k + x_1[2] W_4^{2k} + x_1[3] W_4^{3k}$$

using $W_4 = e^{-j\frac{2\pi}{4}} = (-j)$, we have

$$X_1[k] = 1 + 2(-j)^k + 2(-j)^{2k} + 1(-j)^{3k}$$

$$X_1[0] = 1 + 2 + 2 + 1 = 6$$

$$X_1[1] = 1 + 2(-j) + 2(-j)^2 + (-j)^3 \\ = 1 - 2j - 2 + j = -1 - j$$

$$X_1[2] = 1 - 2 + 2 - 1 = 0$$

$$X_1[3] = 1 + 2(-j)^3 + 2(-j)^4 + 1(-j)^5 = -1 + j$$

similarly $X_2[k]$ can be solved. and $X_3[k] = X_1[k] X_2[k]$

To find $x_3[n]$ we have

$$x_3[n] = \frac{1}{4} \sum_{k=0}^3 X_3[k] W_N^{-kn}$$

$$= \frac{1}{4} [X_3[0] + X_3[1] W_N^{-n} + X_3[2] W_N^{-2n} + X_3[3] W_N^{-3n}]$$

Method IV: Matrix Method:

→ In this method, one sequence is repeated via circular shifting of samples. It is represented as under:

$$x_3[n] = x_1[n] \circledast x_2[n] = x_1[n] \circledast x_1[n]$$

where $\circledast$ indicates circular convolution operation.

$$\begin{bmatrix} x_3[0] \\ x_3[1] \\ x_3[2] \\ \vdots \\ x_3[N-2] \\ x_3[N-1] \end{bmatrix} = \begin{bmatrix} x_2[0] & x_2[N-1] & x_2[N-2] & \dots & x_2[2] & x_2[1] \\ x_2[1] & x_2[0] & x_2[N-1] & \dots & x_2[3] & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & \dots & x_2[4] & x_2[3] \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_2[N-2] & x_2[N-1] & x_2[N-2] & \dots & x_2[0] & x_2[N-1] \\ x_2[N-1] & x_2[N-2] & x_2[N-3] & \dots & x_2[1] & x_2[0] \end{bmatrix} x$$

$$\begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ \vdots \\ x_1[N-2] \\ x_1[N-1] \end{bmatrix}$$

Example: Determine circular convolution of the sequences:

$$x_1[n] = \{1, 2, 2, 1\} \text{ and } x_2[n] = \{2, 1, 1, 2\}$$

→ Solution:

$$\begin{bmatrix} x_3[0] \\ x_3[1] \\ x_3[2] \\ x_3[3] \end{bmatrix} = \begin{bmatrix} x_2[0] & x_2[3] & x_2[2] & x_2[1] \\ x_2[1] & x_2[0] & x_2[3] & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & x_2[3] \\ x_2[3] & x_2[2] & x_2[1] & x_2[0] \end{bmatrix} \begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ x_1[3] \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2+4+2+1 \\ 1+4+4+1 \\ 1+2+4+2 \\ 2+2+2+2 \end{bmatrix} = \begin{bmatrix} 9 \\ 10 \\ 9 \\ 8 \end{bmatrix}$$

$$\therefore x_3[n] = \{9, 10, 9, 8\}$$

- **Linear Convolution Using Circular Convolution:**

➤ The **circular convolution** is given by

$$x_3[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N]$$

Or,
$$x_3[n] = x_3[n] = x_1[n] \circledast x_2[n]$$

➤ Let,

L = length of $x_1[n]$

M = length of $x_2[n]$

then for circular convolution, $L = M$

➤ Also, the length of $x_3[n]$ is same as the length of $x_1[n]$ or $x_2[n]$.

➤ Similarly, the **linear convolution** is given by

$$x_3[n] = \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k]$$

Or,
$$x_3[n] = x_1[n] * x_2[n]$$

➤ Let,

$L = \text{length of } x_1[n]$

$M = \text{length of } x_2[n]$

then the length of $x_3[n]$ (linear convolution) is, $N = L + M - 1$

where, $N = \text{length of } x_3[n]$

Example: Find the linear convolution using circular convolution of the following sequences:

$$x[n] = \{1, 2\} ; h[n] = \{1, 2, 4\}$$

→ Solution:

Given, $x[n] = \{1, 2\}$ and $h[n] = \{1, 2, 4\}$

Then length of $x[n]$, $L = 2$

length of $h[n]$, $M = 3$

Therefore, the length of linear convolution, $N = L + M - 1 = 4$

To make $x[n]$ and $h[n]$ of length 4, $x[n]$ and $h[n]$ are padded with two and one zeros, respectively. Therefore,

$$x[n] = \{1, 2, 0, 0\} \text{ and } h[n] = \{1, 2, 4, 0\}$$

Now, we have,

$$y[n] = \sum_{m=0}^3 x[m] h[(n-m)_4], m=0, 1, 2, 3$$

at $n=0$:

$$y[0] = x[0] h[(0)_4] + x[1] h[(-1)_4] + x[2] h[(-2)_4] + x[3] h[(-3)_4]$$

$$y[0] = x[0] h[0] + x[1] h[3] + x[2] h[2] + x[3] h[1]$$

$$y[0] = 1 \times 1 + 2 \times 0 + 0 \times 4 + 0 \times 2 = 1$$

at $n=1$:

$$y[1] = \sum_{m=0}^3 x[m] h[(1-m)_4]$$

$$y[1] = x[0] h[(1)_4] + x[1] h[(0)_4] + x[2] h[(-1)_4] + x[3] h[(-2)_4]$$

$$y[1] = x[0] h[1] + x[1] h[0] + x[2] h[3] + x[3] h[2]$$

$$y[1] = 1 \times 2 + 2 \times 1 + 0 \times 0 + 0 \times 4 = 4$$

at $n=2$:

$$y[2] = \sum_{m=0}^3 x[m] h[(2-m)_4]$$

$$y[2] = x[0] h[(2-0)_4] + x[1] h[(2-1)_4] + x[2] h[(2-2)_4] + x[3] h[(2-3)_4]$$

$$y[2] = x[0] h[2] + x[1] h[1] + x[2] h[0] + x[3] h[3]$$

$$y[2] = 1 \times 4 + 2 \times 2 + 0 \times 1 + 0 \times 0 = 8$$

at $n=3$:

$$y[3] = \sum_{m=0}^3 x[m] h[(3-m)_4]$$

$$y[3] = x[0] h[(3-0)_4] + x[1] h[(3-1)_4] + x[2] h[(3-2)_4] + x[3] h[(3-3)_4]$$

$$y[3] = x[0] h[3] + x[1] h[2] + x[2] h[1] + x[3] h[0]$$

$$y[3] = 1 \times 0 + 2 \times 4 + 0 \times 2 + 0 \times 1 = 8$$

$$\therefore y[n] = \{1, 4, 8, 8\}$$

This output $y[n]$ obtained is same as those obtained by linear convolution.

Verification:

	$y[0]$	$y[1]$	$y[2]$	$y[3]$
$x[n]$				
1	1	2	4	
2		2	4	8

$$y[0] = 1$$

$$y[1] = 2 + 2 = 4$$

$$y[2] = 4 + 4 = 8$$

$$y[3] = 8$$

6) Additional DFT properties:

6.1) Time reversal of a sequence:

→ This property states that if a sequence is circularly folded, its DFT is also circularly folded. Mathematically,

$$\text{If } x[n] \xrightarrow[N]{\text{DFT}} X[k]$$

$$\text{then } x[(N-n)] \xrightarrow[N]{\text{DFT}} X[(N-k)] = X[N-k]$$

e. Circular Time-Shift:

- It states that shifting the sequences in time-domain by ' m ' samples is equivalent to multiplying the sequence in frequency domain by $e^{-j\frac{2\pi}{N}km}$ or W_N^{km} .
- Mathematically,

$$\begin{aligned} \text{if} \quad & x[n] \xleftrightarrow{DFT} X[k] \\ \text{then} \quad & x[((n-l))_N] \xleftrightarrow{DFT} e^{-j\frac{2\pi}{N}km} X[k] \\ \text{or,} \quad & x[((n-l))_N] \xleftrightarrow{DFT} W_N^{km} X[k] \end{aligned}$$

Proof:

- We know,

Proof: We have

$$\text{DFT}\{x[N-n]\} = \sum_{n=0}^{N-1} x[N-n] e^{-j\frac{2\pi}{N}kn}$$

Let $m = N-n$, then

$$\text{DFT}\{x[N-n]\} = \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}k(N-m)}$$

$$= \sum_{m=0}^{N-1} x[m] e^{j\frac{2\pi}{N}km}$$

$$= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}m(N-k)}$$

$$\text{DFT}\{x[N-n]\} = x[N-k]$$

$$\therefore x[N-n] \xrightarrow{\text{DFT}} x[N-k]$$

Note: Also $\text{DFT}\{x[N-n]\} = \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}(N-k)m} = x[k]$

Also, $x[N-k] = x[(N-k)_N]$ for $0 \leq k \leq N-1$

6.2) Circular time shift of a sequence:

→ This property states that shifting the sequence in time-domain by 'l' samples is equivalent to the multiplying the sequence in frequency domain by $e^{-j\frac{2\pi}{N}kl}$ or W_N^{kl} . Mathematically,

$$x[n] \xrightarrow{\text{DFT}} X[k]$$

$$\text{then } x[(n-l)_N] \xrightarrow{\text{DFT}} X[k] e^{-j\frac{2\pi}{N}kl}$$

Proof: We have

$$\text{DFT}\{x[(n-l)_N]\} = \sum_{n=0}^{N-1} x[(n-l)_N] e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} x[(n-l)_N] e^{-j\frac{2\pi}{N}kn} + \sum_{n=1}^{N-1} x[n-l] e^{-j\frac{2\pi}{N}kn}$$

But $(n-l)_N = N-l+n$

$$\therefore \sum_{n=0}^{N-1} x[(n-l)_N] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} x[N-l+n] e^{-j\frac{2\pi}{N}kn}$$

Let $m = N-l+n$, then

$$\sum_{n=0}^{N-1} x[(n-l)_N] e^{-j\frac{2\pi}{N}kn} = \sum_{m=N-l}^{N-1} x[m] e^{-j\frac{2\pi}{N}k(m-l)}$$

$$= \sum_{m=N-l}^{N-1} x[m] e^{-j\frac{2\pi}{N}k(m-l)} \cdot e^{j\frac{2\pi}{N}kl}$$

$$\text{or } \sum_{n=0}^{N-1} x[(n-1)_N] e^{-j\frac{2\pi}{N}kn} = \sum_{m=N-1}^{N-1} x[m] e^{-j\frac{2\pi}{N}k(m+1)}$$

$$\text{also } \sum_{n=1}^{N-1} x[n-1] e^{-j\frac{2\pi}{N}kn} = \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}k(m+1)}$$

combining we get

$$\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}k(n+1)} = \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}km} e^{-j\frac{2\pi}{N}k}$$

$$= \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}km} e^{-j\frac{2\pi}{N}k} = e^{-j\frac{2\pi}{N}k} \sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}km}$$

$$= e^{-j\frac{2\pi}{N}k} X[k]$$

$$X[(k+1)_N] \xrightarrow{\text{DFT}} X[k] e^{-j\frac{2\pi}{N}k}$$

6.3) Circular Frequency shift:

$$\Rightarrow \text{If } x[n] \xrightarrow{\text{DFT}} X[k] = [X] \times [F] x = [0] \cdot \tilde{0}$$

$$\text{then } x[n] e^{j\frac{2\pi}{N}ln} \xrightarrow{\text{DFT}} X[(k-l)_N]$$

6.4) Multiplication of two sequences:

$$\Rightarrow \text{If } x_1[n] \xrightarrow{\text{DFT}} X_1[k]$$

$$\text{and } x_2[n] \xrightarrow{\text{DFT}} X_2[k]$$

$$\text{then } x_1[n] x_2[n] \xrightarrow{\text{DFT}} \frac{1}{N} X_1[k] \otimes X_2[k]$$

This property is dual of circular convolution property.

6.5) Circular correlation:

→ The circular cross-correlation of two sequences in time-domain is equivalent to the multiplication of DFT of one sequence with the complex-conjugate DFT of other sequence. Mathematically,

If $x[n] \xrightarrow[N]{\text{DFT}} X[k]$

and $y[n] \xrightarrow[N]{\text{DFT}} Y[k]$

then $\tilde{r}_{xy}[l] \xrightarrow[N]{\text{DFT}} \tilde{R}_{xy}[k] = X[k] Y^*[k]$

where, $\tilde{r}_{xy}[l]$ = circular cross-correlation sequence defined as

$$\tilde{r}_{xy}[l] = \sum_{n=0}^{N-1} x[n] y^*[n-l]_N$$

Proof: The circular correlation $\tilde{r}_{xy}[l]$ can be written as

$$\tilde{r}_{xy}[l] = x[l] \circledast y^*[-l]$$

From the circular convolution and complex-conjugate properties, we can write, the N-point DFT of $\tilde{r}_{xy}[l]$ is

$$\tilde{R}_{xy}[k] = X[k] Y^*[k] \quad \left[\because y^*[l] \xrightarrow[N]{\text{DFT}} Y^*[k] \right]$$

If $x[n] = y[n]$, then

$$\tilde{r}_{xx}[l] = x[k] x^*[k] = |x[k]|^2 \xrightarrow[N]{\text{DFT}} \text{energy spectrum}$$

6.6) Parseval's Theorem:

→ For complex-valued sequences $x[n]$ and $y[n]$, if

if $x[n] \xrightarrow[N]{\text{DFT}} X[k]$

and $y[n] \xrightarrow[N]{\text{DFT}} Y[k]$

then $\sum_{n=0}^{N-1} x[n] y^*[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] Y^*[k]$

Proof: We have

$$\tilde{r}_{xy}[0] = \sum_{n=0}^{N-1} x[n] y^*[n]_N$$

then $\tilde{r}_{xy}[0] = \sum_{n=0}^{N-1} x[n] y^*[n] = \sum_{n=0}^{N-1} x[n] y^*[n]$

and also from the definition of IDFT, we have

$$\tilde{X}_y(l) = \frac{1}{N} \sum_{k=0}^{N-1} \{x[k] y^*[k]\} e^{j \frac{2\pi k l}{N}}$$

At $l=0$

$$\tilde{X}_y(0) = \frac{1}{N} \sum_{k=0}^{N-1} x[k] y^*[k]$$

For $x[n] = y[n]$,

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |x[k]|^2$$

7) Relationship of the DFT to Other Transforms:

→ The DFT is an important computational tool for performing frequency analysis of signals on digital signal processors. So, it is important to establish the relationships of the DFT to other transforms.

7.1) Relationship to the Fourier series coefficients of a periodic sequence:

→ The Fourier series of a periodic sequence $x_p[n]$ with fundamental period N is expressed as

$$x_p[n] = \sum_{k=-\infty}^{\infty} c_k e^{j \frac{2\pi k n}{N}}, \quad -\infty < n < \infty$$

where

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p[n] e^{-j \frac{2\pi k n}{N}}, \quad k=0, 1, \dots, N-1$$

is the Fourier series coefficients.

→ We know that the DFT and IDFT is expressed as:

$$X[k] = \sum_{n=0}^{N-1} x_p[n] e^{-j \frac{2\pi k n}{N}}, \quad k=0, 1, \dots, N-1$$

$$\text{and } x_p[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi k n}{N}}, \quad n=0, 1, \dots, N-1$$

→ From the above, we have,

$$X[k] = N c_k$$

and the N -point DFT provides the exact line spectrum of a periodic sequence with fundamental period N .

Fast Fourier Transform (FFT) Algorithm:

1) Introduction:

→ The DFT plays an important role in many applications of digital signal processing, including linear filtering, correlation analysis, and spectrum analysis. A major reason for its importance is the existence of efficient algorithms for computing the DFT.

→ There are two different approaches used in the description of computationally efficient algorithms for evaluating the DFT.

(i) A divide-and-conquer approach:

- In this approach, a DFT of size N , where N is a composite number, is reduced to the computation of smaller DFTs from which the larger DFT is computed.
- This approach presents an important computational algorithm called Fast Fourier Transform (FFT) algorithms, for computing the DFT when the size N is a power of 2 and when it is a power of 4.

(ii) A Linear Filtering Approach

- This approach based on the formulation of the DFT as a linear filtering operation on the data.
- This approach leads to two algorithms, the Goertzel algorithm and the chirp-Z transform algorithm, for computing the DFT via linear filtering of the data sequence.

2) Efficient computation of the DFT: FFT Algorithms.

→ The basic idea behind all fast algorithms for computing the FFT is to decompose successively the N -point DFT computation into computation of smaller size DFTs and to take the advantage of periodicity and symmetry of the complex number W_N^{kn} . If such a decomposition is carried out properly, the algorithm always results in a significant saving in the computational complexity.

→ consider, the DFT of a finite length sequence

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} = \sum_{n=0}^{N-1} x[n] w_N^{kn}, 0 \leq k \leq N-1 \quad \text{--- (1)}$$

$$\text{where, } w_N = e^{-j2\pi/N}$$

→ similarly the IDFT becomes

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] w_N^{-kn}, 0 \leq n \leq N-1 \quad \text{--- (2)}$$

The sequence $x[n]$ is also assumed to be a complex-valued.

→ We observe from equation (1) that direct computation of $X[k]$ requires following number of arithmetic operations:

- (i) N complex multiplications for each value of k .
- (ii) $(N-1)$ complex additions for each value of k .
- (iii) N^2 complex multiplications for N values of k .
- (iv) $N(N-1) = N^2 - N$ complex additions for N values of k .

Note: Since the DFT and IDFT involve basically the same type of computations, the efficient computational algorithms for the DFT applies equally well to the efficient computation of the IDFT.

→ Direct computation of the DFT is basically inefficient because it does not exploit the symmetry and periodicity properties of the phase vector w_N . That is,

$$w_N^{k+N/2} = -w_N^k \quad \text{symmetry property}$$

$$w_N^{k+N} = w_N^k \quad \text{periodicity property}$$

Exploit of these two basic properties results in computational efficient algorithms which are collectively called FFT algorithms.

⇒ Direct computation of the DFT:

→ DFT of a complex-valued sequence $x[n]$ of N -points is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi k}{N} n} \quad (3)$$

where $X[k]$ and $x[n]$ are complex-valued.

$$x[k] = x_R[k] + jx_I[k]$$

$$x_I[n] = x_R[n] + jx_I[n]$$

$$\text{also, } e^{-j \frac{2\pi k}{N} n} = \cos \frac{2\pi k}{N} n + j \sin \frac{2\pi k}{N} n$$

→ Therefore, eqⁿ (3) becomes

$$\begin{aligned} X_R[k] + jX_I[k] &= \sum_{n=0}^{N-1} \{x_R[n] + jx_I[n]\} \left\{ \cos \frac{2\pi k}{N} n - j \sin \frac{2\pi k}{N} n \right\} \\ &= \sum_{n=0}^{N-1} \left\{ x_R[n] \cos \frac{2\pi k}{N} n + x_I[n] \sin \frac{2\pi k}{N} n \right\} \\ &\quad - j \sum_{n=0}^{N-1} \left\{ x_R[n] \sin \frac{2\pi k}{N} n - x_I[n] \cos \frac{2\pi k}{N} n \right\} \end{aligned} \quad (4)$$

→ Representing the real and imaginary parts of eqⁿ (4), we get

$$X_R[k] = \sum_{n=0}^{N-1} \left\{ x_R[n] \cos \frac{2\pi k}{N} n + x_I[n] \sin \frac{2\pi k}{N} n \right\} \quad (5)$$

$$\text{and } X_I[k] = -\sum_{n=0}^{N-1} \left\{ x_R[n] \sin \frac{2\pi k}{N} n - x_I[n] \cos \frac{2\pi k}{N} n \right\} \quad (6)$$

→ The direct computation of eqⁿ (5) and (6) requires:

(i) $2N^2$ evaluations of trigonometric functions.

(ii) $4N^2$ real multiplications ($2N$ by each eqⁿ and $N(2N+2N)=4N^2$)

(iii) $4N(N-1)$ real multiplication, additions, $x[(N-1)-k] = 4N(N-1)$

4) Radix-2 FFT Algorithm:

- The basic strategy used in the FFT algorithm is divide-and-conquer approach, which involves decomposing of an N -point DFT into successively smaller size DFTs.
- This approach is very important when N is highly composite, i.e., when N can be factored as $N = r_1 r_2 r_3 \dots r_v$, where $\{r_j\}$ are prime. For a particular case where $r_1 = r_2 = r_3 = \dots = r_v = r$ so that $N = r^v$. The r is called the radix of the FFT algorithm, and the DFTs are of size r .
- If $r = 2$, then it is called radix-2 FFT algorithm and $N = 2^v$, where v indicates the number of stages in FFT algorithm.
- There are two types of radix-2 FFT algorithms:
 - (a) Decimation-in-Time FFT algorithms (DIT-FFT)
 - (b) Decimation-in-Frequency FFT algorithms (DIF-FFT)

(a) DIT-FFT algorithm:

- The DIT-FFT algorithm is based on decimating (or splitting) $x[n]$ into smaller sequences and finding $X[k]$ from the DFTs of these decimated (or splitted) sequences.
- Let $x[n]$ be a sequence of length $N = 2^v$. Let $x[n]$ is splitted (or decimated) into two subsequences, each of length $\frac{N}{2}$, i.e., even and odd sequences.

$$f_1[n] = x[2n], \quad n = 0, 1, \dots, \frac{N}{2} - 1$$

is an even sequence, and

$$f_2[n] = x[2n+1], \quad 0 \leq n \leq \frac{N}{2} - 1$$

is an odd sequence. also

$$x[n] = f_1[n] + f_2[n]$$

- The N -point DFT of $x[n]$ is

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$

$$\text{or } X[k] = \sum_{n \text{ even}} x[n] W_N^{kn} + \sum_{n \text{ odd}} x[n] W_N^{kn} \quad (3)$$

→ Let $n=2m$ for n even and $n=2m+1$ for n odd, then

$$\begin{aligned} X[k] &= \sum_{m=0}^{N/2-1} x[2m] W_N^{2mk} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{k(2m+1)} \\ &= \sum_{m=0}^{N/2-1} f_1[m] W_N^{2km} + W_N^k \sum_{m=0}^{N/2-1} f_2[m] W_N^{2km} \quad \left[\because W_N^2 = e^{-j\frac{2\pi}{N}2} = e^{-j\frac{2\pi}{N/2}} = W_{N/2} \right] \end{aligned}$$

But $W_N^2 = W_{N/2}$, then

$$X[k] = \sum_{m=0}^{N/2-1} f_1[m] W_{N/2}^{km} + W_N^k \sum_{m=0}^{N/2-1} f_2[m] W_{N/2}^{km} \quad (4)$$

$$\text{or } X[k] = F_1[k] + W_N^k F_2[k], \quad 0 \leq k \leq N-1 \quad (11)$$

where, $F_1[k]$ and $F_2[k]$ are the $\frac{N}{2}$ -point DFTs of the sequences $f_1[m]$ and $f_2[m]$, respectively.

→ $F_1[k + \frac{N}{2}] = F_1[k]$ and $F_2[k + \frac{N}{2}] = F_2[k]$ because $x_1[k]$ and $x_2[k]$ periodic with period $\frac{N}{2}$. Also $W_N^{k+\frac{N}{2}} = -W_N^k$, then eqⁿ (11) becomes

$$\begin{aligned} X[k + \frac{N}{2}] &= F_1[k + \frac{N}{2}] + W_N^{k+\frac{N}{2}} F_2[k + \frac{N}{2}] \\ X[k] &= F_1[k] - W_N^k F_2[k], \quad 0 \leq k \leq \frac{N}{2}-1 \quad (12) \end{aligned}$$

→ A block diagram showing the computations that are necessary for the first stage of an eight-point DIT-FFT is shown below:

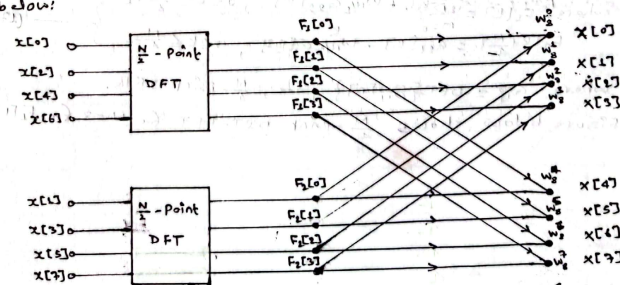


Fig : An eight point DIT-FFT algorithm after the first decimation

→ From above fig we can write

$$x[k] = \begin{cases} F_1[k] + W_N^k F_2[k], & 0 \leq k \leq \frac{N}{2} - 1 \\ F_1[k + \frac{N}{2}] - W_N^k F_2[k + \frac{N}{2}], & \frac{N}{2} \leq k \leq N-1 \end{cases} \quad (13)$$

In the above fig, input sequences are

$$f_1[n] = x[2n] = \{x[0], x[2], x[4], x[6]\}$$

$$\text{and } f_2[n] = x[2n+1] = \{x[1], x[3], x[5], x[7]\}$$

→ If $\frac{N}{2}$ is even, $f_1[n]$ and $f_2[n]$ are again decimated.

Thus $f_1[n]$ would result in the two $\frac{N}{4}$ -point sequences.

$$\left. \begin{aligned} g_{11}[n] &= f_1[2n], & 0 \leq n \leq \frac{N}{4} - 1 \\ g_{12}[n] &= f_1[2n+1], & 0 \leq n \leq \frac{N}{4} - 1 \end{aligned} \right\} \quad (14)$$

and $f_2[n]$ would give

$$\left. \begin{aligned} g_{21}[n] &= f_2[2n], & 0 \leq n \leq \frac{N}{4} - 1 \\ g_{22}[n] &= f_2[2n+1], & 0 \leq n \leq \frac{N}{4} - 1 \end{aligned} \right\} \quad (15)$$

→ By computing $\frac{N}{4}$ -point DFTs, we would obtain the $\frac{N}{2}$ -point DFTs $F_1[k]$ and $F_2[k]$ from the relations

$$\left. \begin{aligned} F_1[k] &= G_{11}[k] + W_{N/2}^k G_{12}[k], & 0 \leq k \leq \frac{N}{4} - 1 \\ F_2[k + \frac{N}{4}] &= G_{11}[k] - W_{N/2}^k G_{12}[k], & 0 \leq k \leq \frac{N}{4} - 1 \end{aligned} \right\} \quad (16)$$

$$\left. \begin{aligned} \text{and } F_2[k] &= G_{21}[k] + W_{N/2}^k G_{22}[k], & 0 \leq k \leq \frac{N}{4} - 1 \\ F_1[k + \frac{N}{4}] &= G_{21}[k] - W_{N/2}^k G_{22}[k], & 0 \leq k \leq \frac{N}{4} - 1 \end{aligned} \right\} \quad (17)$$

where $G_{ij} = \text{DFT}\{g_{ij}[n]\}$, i.e., $\frac{N}{4}$ -point DFTs.

→ Figure below shows $\frac{N}{4}$ -point DIT-FFT for $N=8$ for $F_1[k]$



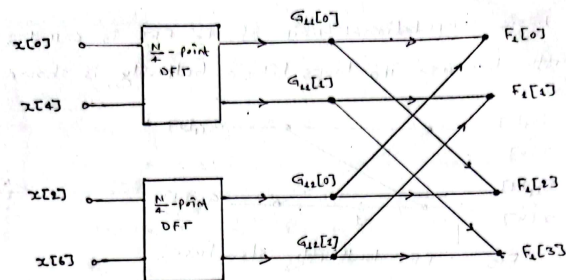


Fig : Decimation of $\frac{N}{4}$ -point (4-point) DFT into two $\frac{N}{4}$ -point (or 2-point) DFTs

→ From Fig we have,

$$G_{11}[n] = G_1[2n] = x[4n] = \{x[0], x[4]\}$$

$$G_{11}[n] = G_1[2n+1] = x[4n+1] = \{x[1], x[5]\}$$

Similar Fig can be obtained for $F_2[k]$. From eqn (17), (18), we have,

$$G_{21}[n] = G_2[2n] = x[4n+2] = \{x[2], x[6]\}$$

$$G_{21}[n] = G_2[2n+1] = x[4n+3] = \{x[3], x[7]\}$$

→ Fig below shows final DIT representation for $N=8$.

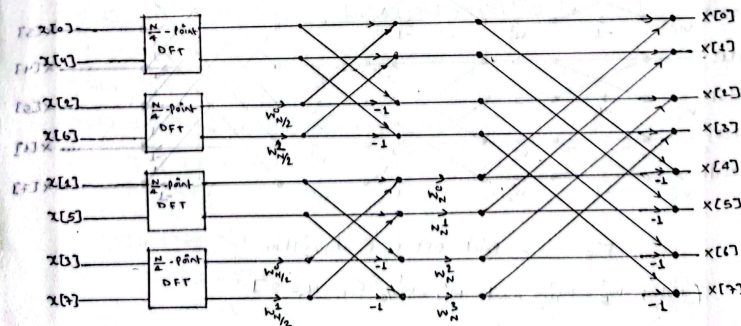


Fig : combined first and second stage of decimation

→ The basic computational unit of the FFT is called a butterfly because it looks like a butterfly, is shown below:

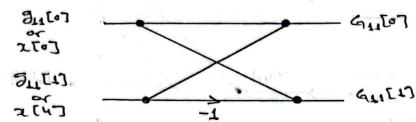


Fig : FFT butterfly structure.

→ We have from definition of DFT

$$G_{11}[k] = \sum_{n=0}^1 x[n] W_N^{kn}, \quad k=0,1$$

$$\therefore G_{11}[0] = \sum_{n=0}^1 x[n] W_N^0 = x[0] + x[1]$$

$$\begin{aligned} \text{and } G_{11}[1] &= \sum_{n=0}^1 x[n] W_N^1 = x[0] W_N^0 + x[1] W_N^1 \\ &= x[0] - x[1] \quad \left[\because W_N^1 = e^{-j\frac{2\pi}{N}} = \cos\theta - j\sin\theta = -1 \right] \end{aligned}$$

→ Therefore, a complete 8-point radix-2 DIT-FFT signal flow graph is shown in Fig below:

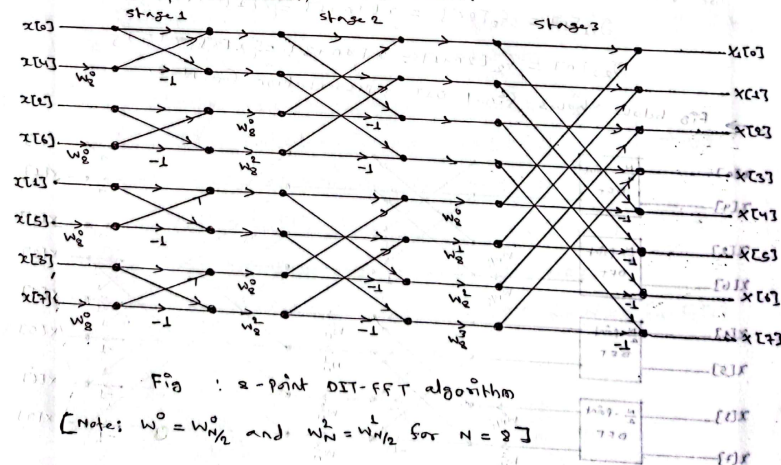


Fig : 8-point DIT-FFT algorithm

[Note: $W_N^0 = W_{N/2}^0$ and $W_N^2 = W_{N/2}^1$ for $N=8$]

Example: Compute the DFT of the sequence
 $x[n] = \{0, 1, 2, 3\}$ using DIT-FFT algorithm.

Solution:

→ Given, $x[n] = \{0, 1, 2, 3\}$

where, $N = 4$ (N is length of sequence $x[n]$)

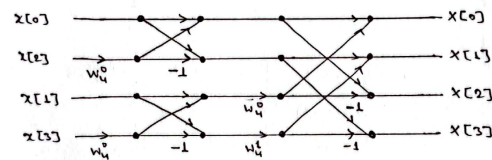
→ The different values of twiddle factor are:

$$W_N = e^{-j\frac{2\pi}{N}}$$
 and for $N = 4$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = -j$$

$$W_4^0 = e^{-j\frac{2\pi}{4} \times 0} = 1$$

→ The 4-point DIT-FFT signal flow graph is shown below:



→ Now, we can find $X[k]$ as follows:

$$X[0] = x[0] + W_4^0 x[2] + [x[1] + W_4^0 x[3]] W_4^0$$

$$= 0 + 2 + [1 + 3]$$

$$\therefore X[0] = 6$$

$$X[1] = -W_4^0 x[2] + x[0] + [W_4^0 x[3] + x[1]] W_4^1$$

$$= -2 + 0 + -j[-3 + 1]$$

$$\therefore X[1] = -2 + j2$$

$$X[2] = -[x[1] + W_4^0 x[3]] W_4^0 + x[0] + W_4^0 x[2]$$

$$= -[1 + 3] + 0 + 2$$

$$= -4 + 2$$

$$\therefore X[2] = -2$$

$$X[3] = -[x[2] - W_4^0 x[3]] W_4^1 + x[0] - W_4^0 x[2]$$

$$= -[2 - 3](-j) + 0 - 2$$

$$= -2j - 2$$

$$\therefore X[3] = -2 - 2j$$

$$\therefore X[k] = \{6, -2 + j2, -2, -2 - 2j\}$$

- **Decimation in Frequency FFT (DIF-FFT) Algorithm: (Assignment)**